

Natalie Leaist

647-248-8430 | nleaist@student.ubc.ca | Natalie-Leaist.github.io | linkedin.com/in/natalie-leaist/ | github.com/Natalie-Leaist

EDUCATION

University of British Columbia

Vancouver, BC, Canada

Bachelor of Applied Science in Engineering Physics, minor in Honours Math

September 2024 – Present

- 94% cumulative average (4.33/4.33 GPA), 96% average across math and physics coursework
- Trek Excellence Scholarship (top 5% of faculty and year), Thomas Beeching Scholarship, EnSURE Award & Dean's Scholar

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, JavaScript

Data Analysis and Modelling: FFT/Spectral Analysis, basic statistical tests, NumPy, matplotlib, Jupyter, MATLAB, Simulink

Embedded Systems: Microcontrollers (STM32, ESP32, TMS570), FreeRTOS, Communication Protocols, ADCs, Sensors, Arduino

Electrical: Circuit Design, PCB Design, KiCad, Oscilloscope, Logic Analyzer, Function Generator, Soldering

Other Tools: Git, Linux, VSCode, OpenCV, L^AT_EX

Machining: 35 Hours of training in machining strategies and techniques, including mill, lathe, waterjet cutter, drill press and hand tools

EXPERIENCE

R&D Electronic Design Intern

January 2026 – April 2026

Blue Robotics

Victoria, BC

- Designed and tested a switch-mode power supply development board featuring buck and SEPIC (Single-Ended Primary-Inductor) converter topologies, including inductor ripple current calculations, transfer function analysis, and compensation network design
- Engineered a 100A power distribution board powering up to 8 ESCs (electronic speed controllers) with ethernet communication, designing multi-layer copper pour layouts and specifying 2oz copper to prevent thermal and reliability failures at high current
- Developed a compact (2x3 cm) data logger PCB centered around an ATmega32u4 MCU, including power multiplexing between battery and USB-C, battery supervision, a buck-boost converter, SD card logging as well as analog signal filtering and buffering
- Evaluated closed-loop behavior using Bode plots and gain/phase margin analysis

Research Intern & Software Developer

May 2025 – August 2025

Experimental Cosmology Group, UBC Physics & Astronomy

Vancouver, BC

- Worked on the CGEM project, a radio telescope that will be used to help detect gravitational radiation from the early universe (collaboration between UBC, NASA Goddard Space Flight Centre and DRAO)
- Developed a high-frequency (830 Hz) embedded data acquisition system for accelerometer-based vibration and inclination monitoring of the telescope using an STM32 microcontroller, a 32-bit sigma-delta ADC, DMA, SPI communication, FreeRTOS, and Ethernet streaming
- Performed Python-based FFT, spectral noise analysis, and calibration, to verify accelerometer performance and achieve less than 1 μm displacement resolution
- Optimized embedded timing, memory alignment, and signal integrity, improving system reliability and throughput

Control Systems & Software Developer

September 2024 – Present

UBC Orbit Satellite Design Team

Vancouver, BC

- Working on the ALEASAT project, an Earth observation satellite supported by the European Space Agency's Fly Your Satellite! Program
- Designed a Helmholtz cage to simulate the magnetic field of low Earth orbit
- Presented findings on Helmholtz cage with a poster and presentation at an undergraduate physics conference (CCUW*IP 2026)
- Developing Python simulations of satellite attitude dynamics and control laws, assisting in designing and validating ALEASAT's attitude determination and control systems
- Constructed firmware and software to fulfill mission and testing requirements set by the European Space Agency, including CPU usage monitoring across FreeRTOS tasks

Outdoor Adventure Guide

July 2022 – August 2023

The Adventure Group

Whistler, BC

- Led groups of up to 20 guests on Canada's highest and 2nd longest zip-line
- Received over 50 five-star guest reviews, the highest among guides, highlighting strong communication and leadership
- Conducted safety inspections on off-road vehicles, zip-lines, and aerial obstacle courses
- Trained in leadership, emergency response, risk assessment, and guest management

PROJECTS

Self-Balancing Robot

- Designed and built a two-wheeled self-balancing robot using Arduino, MPU6050 IMU, DRV8825 motor drivers, and NEMA- 17 stepper, gaining hands-on experience in hardware integration and embedded programming
- Implemented a complementary filter to combine accelerometer and gyroscope data, providing a stable pitch estimate and developing practical skills in sensor fusion and noise reduction
- Developed and tuned a PID controller to actively stabilize the robot's pitch

Wordle Solver

- Developed a Python algorithm capable of solving Wordle puzzles with an average of approximately 4 guesses
- Implemented a word-elimination strategy to reduce the number of possible solutions after each guess
- Used object-oriented programming in Python to organize the solver into modular, reusable classes

ACTIVITIES

Freeride Skiing

- Ranked 2nd in Canada (2022-2023 season), qualifying for Junior World Championships (2023) as one of 15 junior female skiers worldwide
- Recognized with Rossignol Freeride Platinum Sponsorship (Sept. 2023 – Aug. 2024), the highest athlete tier before Rossignol's professional team